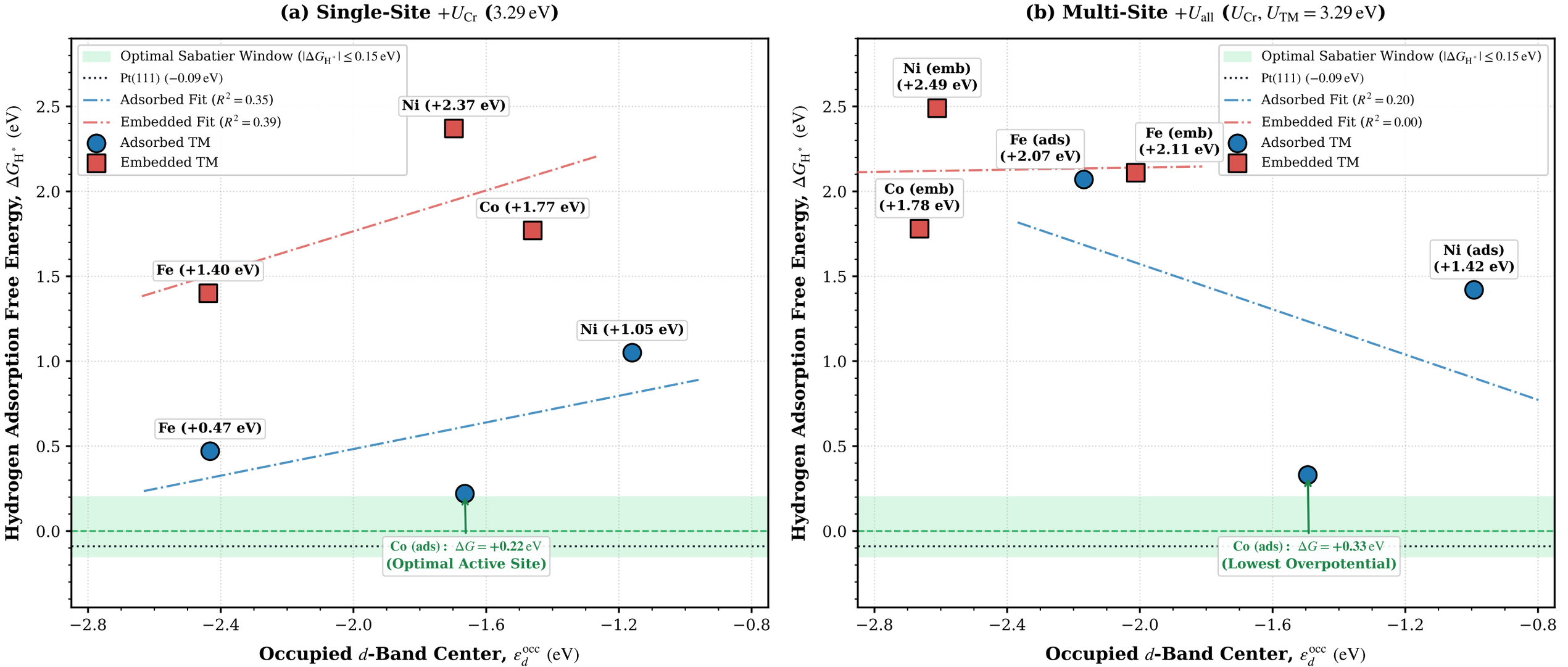

(a) Single-Site +U_{Cr} (3.29 eV)
Optimal Sabatier Window ($|\Delta G_{H^*}| \leq 0.15$ eV)
Pt(111) (−0.09 eV)
Adsorbed Fit ($R^2 = 0.35$)
Embedded Fit ($R^2 = 0.39$)
Adsorbed TM
Embedded TM
Ni (+2.37 eV)
Co (+1.77 eV)
Fe (+1.40 eV)
Ni (+1.05 eV)
Fe (+0.47 eV)
Co (ads) : $\Delta G = +0.22$ eV
(Optimal Active Site)
Hydrogen Adsorption Free Energy, ΔG_{H^*} (eV)
Occupied d-Band Center, ε_d^{occ} (eV)
(b) Multi-Site +U_{all} (U_{Cr}, $U_{TM} = 3.29$ eV)
Optimal Sabatier Window ($|\Delta G_{H^*}| \leq 0.15$ eV)
Pt(111) (−0.09 eV)
Adsorbed Fit ($R^2 = 0.20$)
Embedded Fit ($R^2 = 0.00$)
Adsorbed TM
Embedded TM
Ni (emb)
(+2.49 eV)
Fe (ads)
(+2.07 eV)
Fe (emb)
(+2.11 eV)
Co (emb)
(+1.78 eV)
Ni (ads)
(+1.42 eV)
Co (ads) : $\Delta G = +0.33$ eV
(Lowest Overpotential)
Hydrogen Adsorption Free Energy, ΔG_{H^*} (eV)
Occupied d-Band Center, ε_d^{occ} (eV)